

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0253

Brush Tavern 138 kV
April 26, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0253. The Interconnection Customer proposes a 446.4 MW wind facility interconnecting at the Brush Tavern 138 kV substation in the Virginia area. The request was received on July 04, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 158 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,407,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Intel Substation - S Creek 138 KV
Contingency: S Crk - Aibonito Pds 138 KV
Loading: 115.3% (Rating: 191 MVA)
DFAX: 6.8%
Pre-project loading: 99.4%
2. Facility: Louisa Generation Facility 345 kV/138 kV Auto
Contingency: Louisa Generation Facility Substation - Fluvanna P.S 345 KV L
Loading: 113.1% (Rating: 462 MVA)
DFAX: 6.3%
Pre-project loading: 107.0%
3. Facility: Carmel 146Th St - Oak Crk 138 KV
Contingency: Clver - Bear 345 KV
Loading: 111.2% (Rating: 156 MVA)
DFAX: 5.4%
Pre-project loading: 95.7%
4. Facility: S Crk - Wythe 138 KV
Contingency: Candler's Mountain - Prkin 138 KV L
Loading: 109.4% (Rating: 255 MVA)
DFAX: 9.7%
Pre-project loading: 92.4%
5. Facility: Ceiba Baja 7012 - Carmel 146Th St 138 KV
Contingency: Clover - Bear 345 KV
Loading: 107.8% (Rating: 164 MVA)
DFAX: 5.4%
Pre-project loading: 93.1%

6. Facility: Bear - Leavitt 345 KV Ckt 1
 Contingency: Bear - Leavitt Substation 345 KV #2
 Loading: 104.7% (Rating: 1,580 MVA)
 DFAX: 15.2%
 Pre-project loading: 100.5%
7. Facility: Rustburg Substation - Monel Substation 138 KV
 Contingency: Bear - Clover 345 KV
 Loading: 103.3% (Rating: 355 MVA)
 DFAX: 39.1%
 Pre-project loading: 54.1%
8. Facility: Brush Tavern Substation - Rustbug 138 KV line
 Contingency: Bear - Clover 345 KV Line
 Loading: 101.7% (Rating: 393 MVA)
 DFAX: 39.1%
 Pre-project loading: 57.4%
9. Facility: Monel - Candelaria 138 KV
 Contingency: Bear - Clover 345 KV
 Loading: 101.7% (Rating: 379 MVA)
 DFAX: 11.7%
 Pre-project loading: 87.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Crk - Intel 138 KV line	Rebuild line with higher-capacity conductor	\$7,332,000	21% (\$1,530,000)
Louisa Generation Facility 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$23,482,000	14% (\$3,263,000)
Carmel 146Th St - Oak Creek 138 KV	Rebuild line with higher-capacity conductor	\$4,344,000	56% (\$2,451,000)
S Crk - Wythe 138 KV	Reconductor line	\$4,510,000	18% (\$814,000)
Ceiba Baja 7012 - Carmel 146Th St 138 KV	Reconductor line	\$6,099,000	56% (\$3,441,000)
Bear - Leavitt 345 KV Ckt 1	Reconductor line	\$62,537,000	9% (\$5,628,000)
Monel - Rustburg 138 KV	Reconductor line	\$3,149,000	57% (\$1,792,000)
Brush Tavern Substation - Rustbug 138 KV line	Reconductor line	\$2,555,000	57% (\$1,454,000)
Monel - Candelaria 138 KV	Reconductor line	\$9,460,000	22% (\$2,034,000)
	Total allocated to project		\$22,407,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.